Appendix D: Analytical calculations of A_{11}, A_{22}, A_{12}, D_{11}, D_{22} for each laminate

Steps:

1. Find material properties of composite based on fiber/matrix properties (from Lab 1 calculations)
2. Find Q in material axes based on material properties
3. Find $\bar{Q}$ at 0°, 90°, 45°, and -45°
4. Find [A] and [D] for the 2 layups:
   (a) [0/90/90/0]
   (b) [45/0/0/45]

Step 1: Composite material properties

From Lab 1, the theoretical values of the composite's material properties in the material axes are:

$E_1 = 139.5$ GPa $\nu_{21} = 0.0120$

$E_2 = 6.101$ GPa $G_1 = 2.574$ GPa

$\nu_{12} = 0.2750$

Step 2: Find Q

$$Q = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{33} \end{bmatrix}$$

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}} \qquad Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}} \qquad Q_{33} = G_{12}$$

$$Q_{12} = Q_{21} = \frac{\nu_{12} E_2}{1-\nu_{12}\nu_{21}}$$

Plug values into Matlab:

$$Q = \begin{bmatrix} 140.0 & 1.683 & 0 \\ 1.683 & 6.121 & 0 \\ 0 & 0 & 2.574 \end{bmatrix} \text{GPa}$$

Step 3: Find $\bar{Q}$ at 0°, 90°, 45°, -45°

Transformation matrix: $T^{-1} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \cos\theta\sin\theta & -\cos\theta\sin\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$, then $\bar{Q} = T^{-1} Q T^{-T}$

Do this in Matlab

$$\bar{Q}^0 = Q = \begin{bmatrix} 140.0 & 1.683 & 0 \\ 1.683 & 6.121 & 0 \\ 0 & 0 & 2.574 \end{bmatrix} \text{GPa}$$

$$\bar{Q}^{90} = \begin{bmatrix} 6.121 & 1.683 & 0 \\ 1.683 & 140.0 & 0 \\ 0 & 0 & 2.574 \end{bmatrix} \text{GPa}$$

$$\bar{Q}^{45} = \begin{bmatrix} 39.94 & 34.79 & 33.46 \\ 34.79 & 39.94 & 33.46 \\ 33.46 & 33.46 & 35.68 \end{bmatrix} \text{GPa}$$

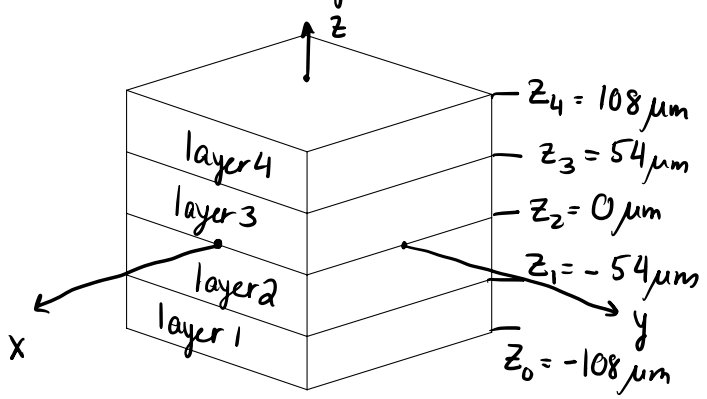
$$\bar{Q}^{-45} = \begin{bmatrix} 39.94 & 34.79 & -33.46 \\ 34.79 & 39.94 & -33.46 \\ -33.46 & -33.46 & 35.68 \end{bmatrix} \text{GPa}$$

Step 4: Find [A] and [D]

$$[A] = \sum_{k=1}^{n} \bar{Q}^k (z_k - z_{k-1}) \qquad [D] = \sum_{k=1}^{n} \bar{Q}^k \left(\frac{z_k^3 - z_{k-1}^3}{3} \right)$$

(n: number of layers)

Each laminate has 4 layers, each of thickness 54 μm



$$[A] = \bar{Q}^1 \underbrace{(z_1 - z_0)}_{=54\,\mu m} + \bar{Q}^2 \underbrace{(z_2 - z_1)}_{=54\,\mu m} + \bar{Q}^3 \underbrace{(z_3 - z_2)}_{=54\,\mu m} + \bar{Q}^4 \underbrace{(z_4 - z_3)}_{=54\,\mu m}$$

$$[A] = 54\,\mu m \left(\bar{Q}^1 + \bar{Q}^2 + \bar{Q}^3 + \bar{Q}^4 \right)$$

$$[D] = \bar{Q}^1 \left(\frac{z_1^3 - z_0^3}{3} \right) + \bar{Q}^2 \left(\frac{z_2^3 - z_1^3}{3} \right) + \bar{Q}^3 \left(\frac{z_3^3 - z_2^3}{3} \right) + \bar{Q}^4 \left(\frac{z_4^3 - z_3^3}{3} \right)$$

Run these in Matlab for the 2 laminates

[A] has units of GPa·μm : to convert to Pa·m, multiply by $\left(\frac{10^9 \text{ Pa}}{1 \text{ GPa}}\right)\left(\frac{1 \text{ m}}{10^6 \mu m}\right) = 10^3$

[D] has units of GPa·μm^3 : to convert to Pa·m^3, multiply by $\left(\frac{10^9 \text{ Pa}}{1 \text{ GPa}}\right)\left(\frac{1 \text{ m}}{10^6 \mu m}\right)^3 = 10^{-9}$

(a) [0/90/90/0]

$$[A] = \begin{bmatrix} 1.578 & 0.03636 & 0 \\ 0.03636 & 1.578 & 0 \\ 0 & 0 & 0.05560 \end{bmatrix} \times 10^4 \text{ GPa} \cdot \mu\text{m}$$

$$[A] = \begin{bmatrix} 1.578 & 0.03636 & 0 \\ 0.03636 & 1.578 & 0 \\ 0 & 0 & 0.05560 \end{bmatrix} \times 10^7 \underbrace{\text{Pa} \cdot \text{m}}_{= \text{N/m}}$$

$A_{11} = 1.578 \times 10^7$ N/m
$A_{12} = 0.03636 \times 10^7$ N/m
$A_{22} = 1.578 \times 10^7$ N/m
for [0/90/90/0]

$$[D] = \begin{bmatrix} 1.035 & 0.01414 & 0 \\ 0.01414 & 0.1919 & 0 \\ 0 & 0 & 0.02162 \end{bmatrix} \times 10^8 \text{ GPa} \cdot \mu\text{m}^3$$

$$[D] = \begin{bmatrix} 1.035 & 0.01414 & 0 \\ 0.01414 & 0.1919 & 0 \\ 0 & 0 & 0.02162 \end{bmatrix} \times 10^{-1} \underbrace{\text{Pa} \cdot \text{m}^3}_{= \text{N} \cdot \text{m}}$$

$D_{11} = 0.1035$ N·m
$D_{22} = 0.01919$ N·m
for [0/90/90/0]

(b) [45/0/0/45]

$$[A] = \begin{bmatrix} 1.943 & 0.3939 & 0.3614 \\ 0.3939 & 0.4974 & 0.3614 \\ 0.3614 & 0.3614 & 0.4131 \end{bmatrix} \times 10^4 \text{ GPa} \cdot \mu\text{m}$$

$$[A] = \begin{bmatrix} 1.943 & 0.3939 & 0.3614 \\ 0.3939 & 0.4974 & 0.3614 \\ 0.3614 & 0.3614 & 0.4131 \end{bmatrix} \times 10^7 \underbrace{\text{Pa} \cdot \text{m}}_{= \text{N/m}}$$

$A_{11} = 1.943 \times 10^7$ N/m
$A_{12} = 0.3939 \times 10^7$ N/m
$A_{22} = 0.4974 \times 10^7$ N/m
for [45/0/0/45]

$$[D] = \begin{bmatrix} 4.404 & 2.574 & 2.459 \\ 2.574 & 2.999 & 2.459 \\ 2.459 & 2.459 & 2.649 \end{bmatrix} \times 10^{7} \ GPa \cdot \mu m^3$$

$$[D] = \begin{bmatrix} 4.404 & 2.574 & 2.459 \\ 2.574 & 2.999 & 2.459 \\ 2.459 & 2.459 & 2.649 \end{bmatrix} \times 10^{-2} \ \underbrace{Pa \cdot m^3}_{= N \cdot m}$$

$D_{11} = 4.404 \times 10^{-2} \ N \cdot m$

$D_{22} = 2.999 \times 10^{-2} \ N \cdot m$

for [45/0/0/45]